# 24.1 Ultrasound & X-rays

## Question Paper

| | |
|---|---|
| Course | CIE A Level Physics |
| Section | 24. Medical Physics |
| Topic | 24.1 Ultrasound & X-rays |
| Difficulty | Hard |

**Time allowed:** 70

**Score:** /53

**Percentage:** /100

## Question 1a

Most modern ultrasound transducers are made from the piezoelectric material lead zirconate titanate (PZT).

Fig. 1.1 shows the effect on the structure of a crystal of PZT when an alternating potential difference is applied across it.

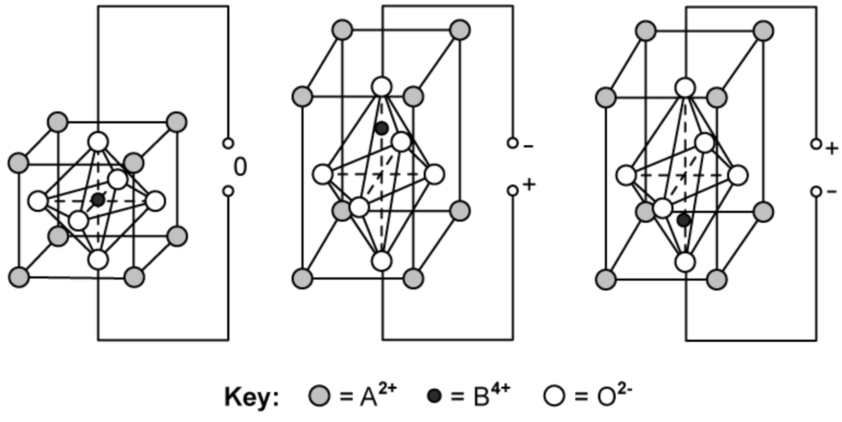



**Fig. 1.1**

(i)

Describe what happens at a molecular level when an alternating potential difference is applied across a PZT transducer to generate ultrasound.

You may draw on Fig. 1.1 to help with your answer.

[4]

(ii)

Describe the conditions that will enable maximum energy conversion into ultrasound to occur.

[2]

**[6 marks]**

## Question 1b

An eye can be imaged using either an A-scan or a B-scan ultrasound. Fig. 1.2 shows the position of a piezoelectric transducer being used during an A-scan of an eye.

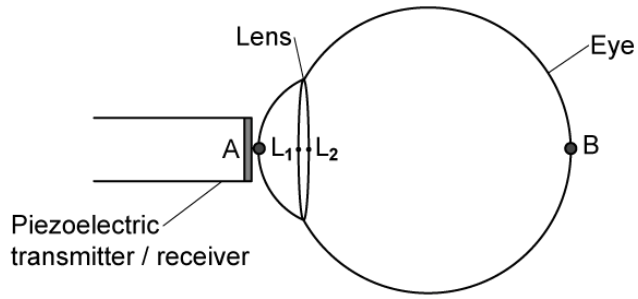



**Fig. 1.2**

(i)

Explain how an A-scan could be used to measure the thickness of a patient's eye lens. You may draw on Fig. 1.2 to help with your answer.

[3]

(ii)

Explain why a B-scan would be better than an A-scan for imaging damaged tissue that surrounds the eye.

[2]

**[5 marks]**

## Question 1c

Fig. 1.3 shows an ultrasound A-scan trace from the scan of an eye.

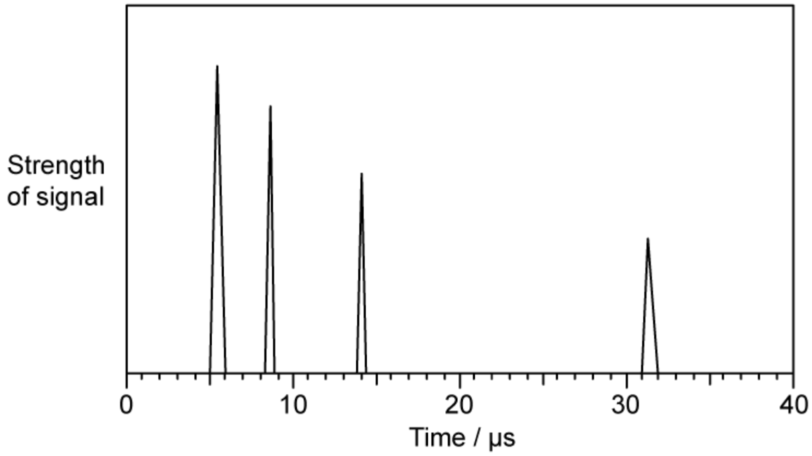



**Fig. 1.3**

The speed of ultrasound in the eye is 1520 m $s^{-1}$ and the speed of ultrasound in the lens is 1670 m $s^{-1}$.

Use Fig. 1.3 to calculate the thickness of

(i)

the lens, in mm,

[2]

(ii)

the eye, in cm.

[3]

**[5 marks]**

## Question 1d

The graph in Fig. 1.4 shows the attenuation of the intensity of ultrasound at a frequency of 3 MHz with depth.

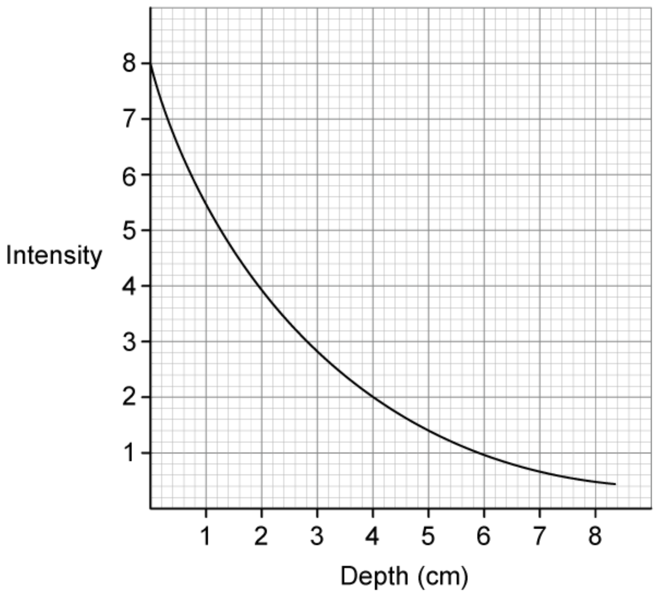



**Fig. 1.4**

When carrying out the scan, a gel is applied between the transducer and eye to enable 100% transmission of the ultrasound into the eye.

Use the information in Table 1.1 and the graph in Fig. 1.4 to calculate the ratio of the intensity of the reflections from A and B, by the time the pulses come back to the receiver.

**Table 1.1**

| | **acoustic impedance / kg $m^{-2}$ $s^{-1}$** |
|---|---|
| eye | $1.52 \times 10^6$ |
| eye lens | $1.84 \times 10^6$ |
| surrounding tissue | $1.69 \times 10^6$ |

**[4 marks]**

## Question 2a

The data in Table 1.1 shows how the attenuation coefficient $\mu$ depends on the energy of the X-rays in bone and muscle.

**Table 1.1**

| maximum X-ray energy / keV | $\mu_{bone}$ / $cm^{-1}$ | $\mu_{muscle}$ / $cm^{-1}$ |
|---|---|---|
| 50 | 3.32 | 0.54 |
| 100 | 0.60 | 0.21 |
| 250 | 0.32 | 0.16 |
| 4000 | 0.087 | 0.049 |

Using the data in Table 1.1, state and explain which X-ray energy would produce an image of a bone next to a muscle with the best contrast.

**[3 marks]**

## Question 2b

For X-ray energies around 30 keV and below, the attenuation coefficient $\mu$ is approximately proportional to the cube of the proton number $Z$ of the absorbing material.

The proton numbers of muscle, bone and some contract media are shown in Table 1.2

**Table 1.2**

| substance | main elements | |
|---|---|---|
| muscle | ${}_{1}H$, ${}_{6}C$, ${}_{8}O$ | |
| bone | ${}_{1}H$, ${}_{6}C$, ${}_{8}O$, ${}_{15}P$, ${}_{20}Ca$ | |
| contrast media | ${}_{53}I$, ${}_{56}Ba$ | |

Using the information in Table 1.2

(i)
Show that bone absorbs X-rays about eight times more strongly than muscle.

[3]

(ii)
Explain why contrast media are made from elements with high values of atomic number $Z$.

[3]

**[6 marks]**

## Question 2c

The graph in Fig. 1.1 shows how the attenuation coefficient $\mu$ of muscle varies with photon energy $E$.

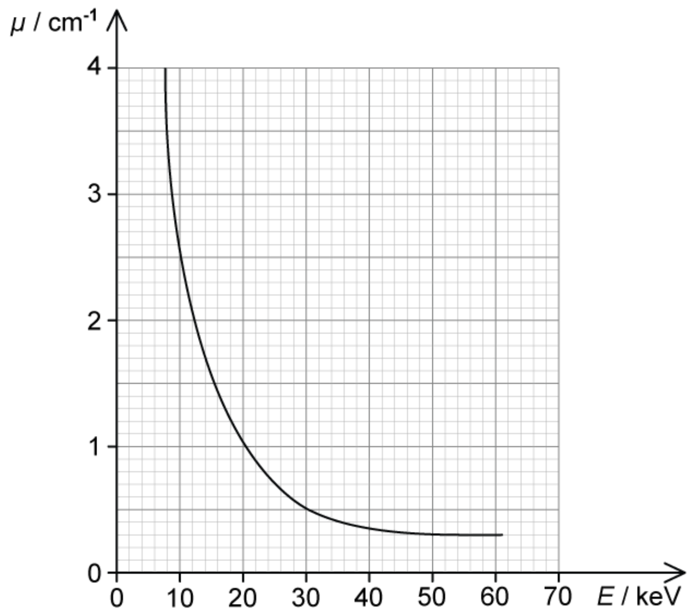



**Fig. 1.1**

Using Fig. 1.1, explain why X-rays with energies less than 20 keV are usually filtered out of the beam.

**[4 marks]**

## Question 2d

An X-ray beam of energy 20 keV and intensity $I_0$ is incident on a sample of muscle and bone as shown in Fig. 1.2. The emergent beam intensities from the muscle and bone are $I_m$ and $I_b$ respectively.

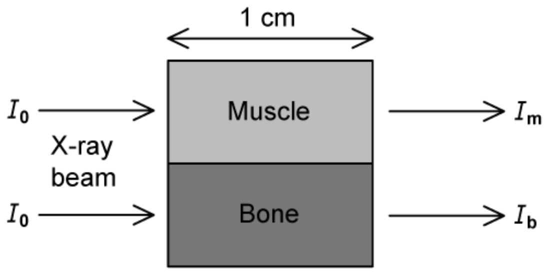



**Fig. 1.2**

Using the graph in Fig. 1.1, calculate the ratio of intensities to compare the attenuation produced by 1 cm of bone and 1 cm of muscle.

**[3 marks]**

## Question 3a

An ultrasound investigation was used to identify a small volume of an unknown substance in a patient. It is suspected that this substance is either a cyst or a tumour.

During the ultrasound investigation, an ultrasound pulse of frequency 5 MHz is passed through soft tissue and then into the small volume of the unknown substance.

The ratio of the reflected intensity to the incident intensity for the ultrasound pulse reflected at the boundary was found to be $7.54 \times 10^{-5}$.

Table 1.1 shows some information about the density and speed of ultrasound in soft tissue, cysts and tumours.

**Table 1.1**

| material | density / kg $m^{-3}$ | ultrasound velocity / m $s^{-1}$ |
|---|---|---|
| soft tissue | 1065 | 1530 |
| cyst | 1020 | 1570 |
| tumour | 990 | 1565 |

(i)

Use the information in Table 1.1 to show that the unknown medium is a cyst.

[4]

(ii)

A pulse of ultrasound reflected from the front surface of the cyst was detected 27.5 μs later and a pulse from the rear surface was detected 43.2 μs later.

Determine the length of the cyst, in cm.

[2]

**[6 marks]**

## Question 3b

Fig. 1.1 shows a beam of X-rays of energy 20 keV incident normally on some soft tissue which contains the cyst.

The cyst is located 2.1 cm from the surface of the soft tissue.

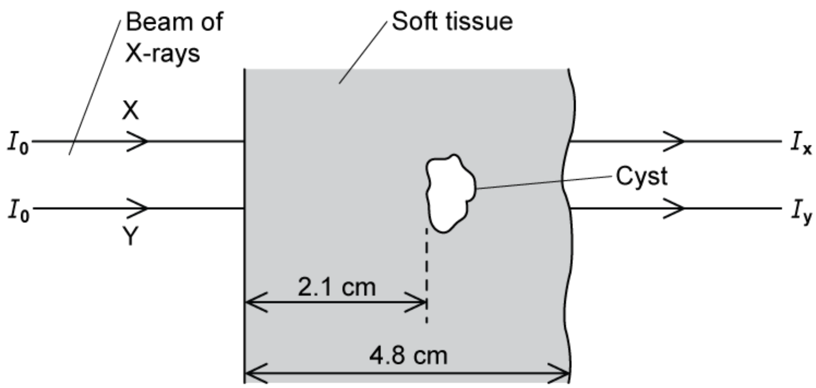



**Fig. 1.1** (not to scale)

The attenuation constant of the soft tissue is 0.85 cm$^{-1}$. The incident intensity of the beam is $4.6 \times 10^3$ W m$^{-2}$.

The X-ray beam is switched on for a time *t*.

Determine the energy of the X-ray beam incident on the front surface of the cyst each second. State any assumptions you make.

**[3 marks]**

## Question 3c

The ratio of the intensities of the emergent X-ray beams X and Y is determined to be

$$\frac{I_Y}{I_X} = 0.94$$

(i)
Determine the attenuation coefficient of the cyst.

[3]

(ii)
Compare the quality of the images that would be obtained from the ultrasound and the X-ray scans of the cyst.

[2]

**[5 marks]**

## Question 3d

Suggest a different imaging technique that could be used to image the cyst. Explain **one** advantage and **one** disadvantage of using this technique over ultrasound.

**[3 marks]**

# Model Answers: Hard

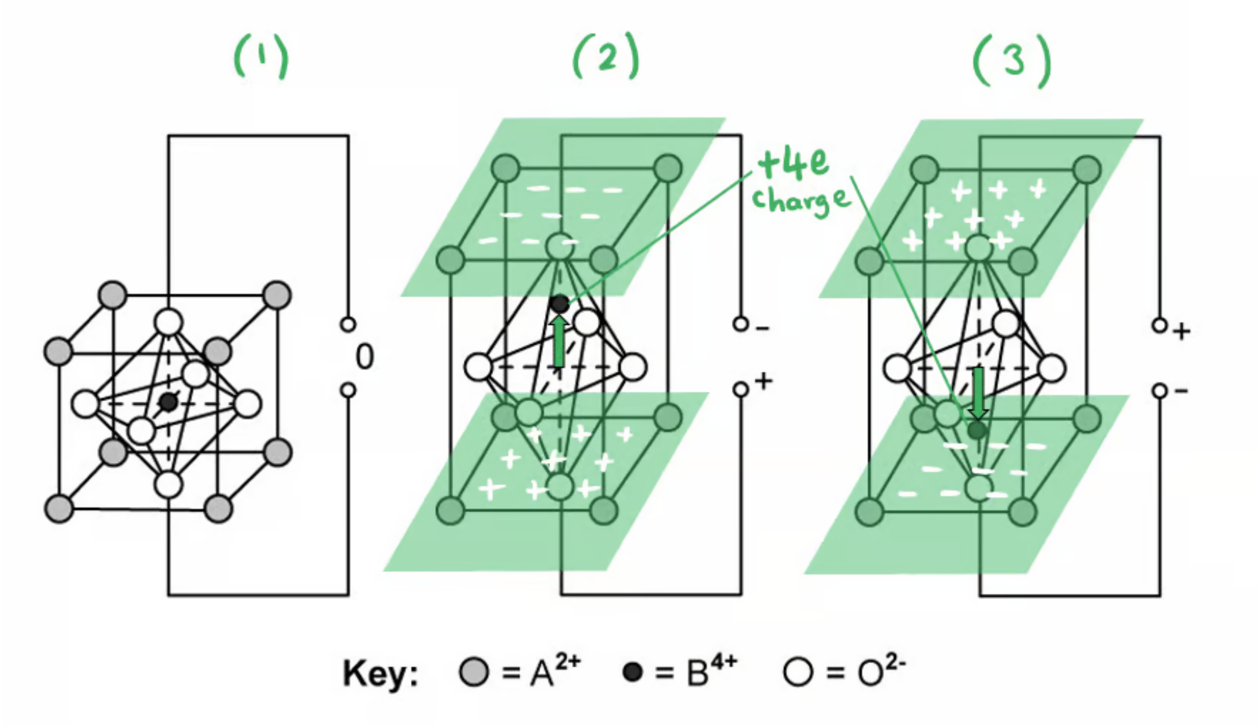
1a

(a)

(i) When an alternating potential difference (p.d.) is applied to a PZT transducer:

Any **four** from:

- The positive / negative charges are attracted to the negative / positive terminal **OR** charges in the lattice (of the PZT crystal) separate; [1 mark]
- The movement of the central positive charge (along the central axis) causes the PZT crystal to deform / change shape / expand and contract; [1 mark]
- When the p.d. changes direction, the central positive charge moves in the opposite direction, causing the crystal to deform (in the opposite direction); [1 mark]
- The repeated crystal deformations lead to oscillations **OR** demonstrate on diagram e.g. $(1) \xrightarrow{expanding} (2) \xrightarrow{contracting} (1) \xrightarrow{expanding} (3) \xrightarrow{contracting} (1)$ [1 mark]
- The alternating p.d. causes the crystal to oscillate at the same frequency (as the applied field); [1 mark]
- The oscillations are transferred to the surrounding air molecules to produce ultrasound (if frequency > 20 000 Hz); [1 mark]



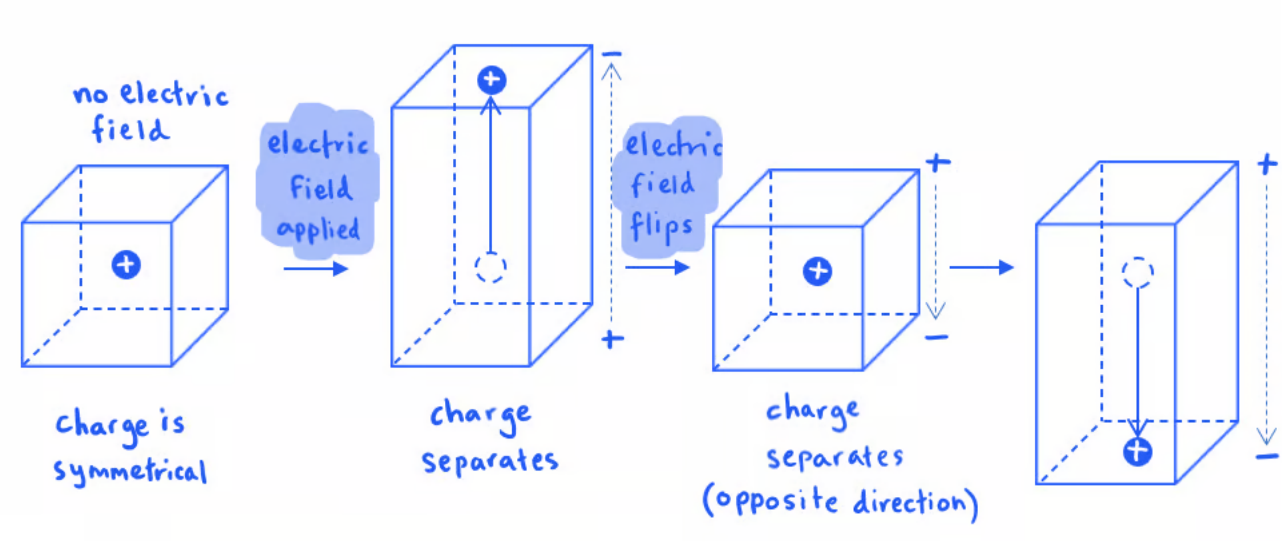

(ii) Maximum energy transfer occurs when:

- The frequency of the applied (alternating) p.d. is equal to the natural frequency of the PZT crystal / transducer; [1 mark]
- This causes resonance to occur; [1 mark]

**[Total: 6 marks]**

The diagram might look daunting at first, but the key part to notice is the **movement of the central ion** which has a net charge of +4e

A simplified view is shown below:




1b

(b)

(i) An A-scan can be used to measure the thickness of a patient's eye lens by:

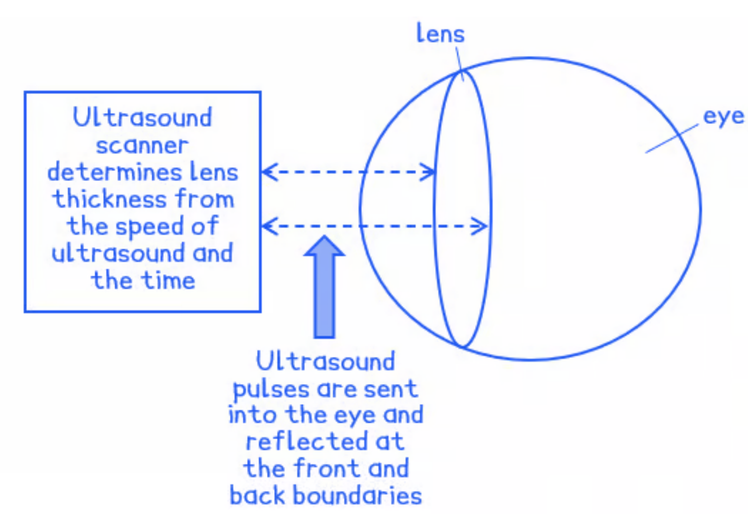
- Pulses of ultrasound are sent into the eye; [1 mark]
- The front and back of the lens reflect the pulses (which are displayed on an oscilloscope); [1 mark]
- The thickness of the lens is determined from the speed of ultrasound (in the medium) and the time; [1 mark]

(ii) A B-scan is better for imaging damaged tissue that surrounds the eye because:

- A-scans are one-directional, they can only be used to take measurements of the eye; [1 mark]
- B-scans produce 2 or 3-dimensional images, so they can be used to investigate the structures in / around the eye; [1 mark]

**[Total: 5 marks]**

The paths of the ultrasound pulses are as follows:



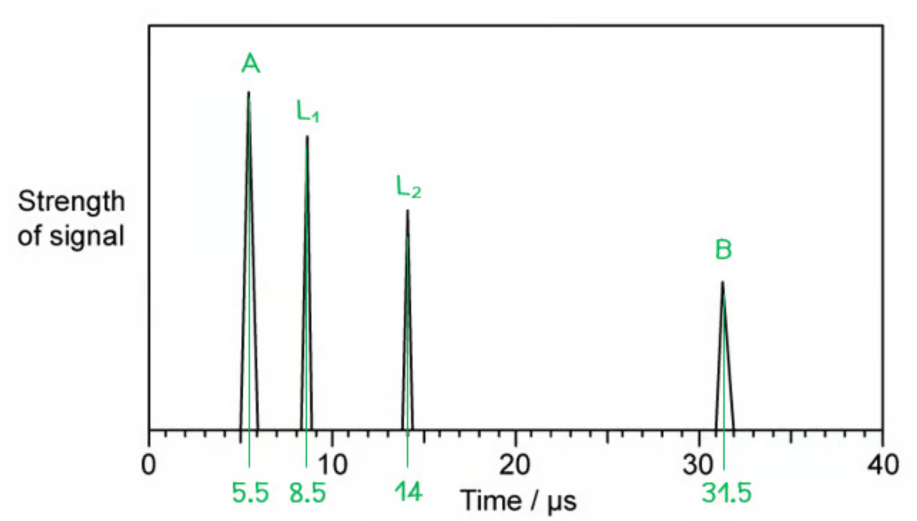

1c

(c) Calculate the thickness of the eye lens:

List the known quantities:

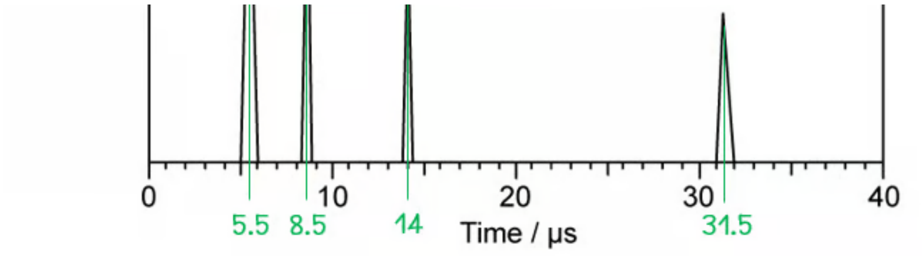
- Speed of ultrasound in the eye, $c_{eye}$ = 1520 m s$^{-1}$
- Speed of ultrasound in the lens, $c_{lens}$ = 1670 m s$^{-1}$

(i) Calculate the thickness of the lens:




- (From Fig. 1.3) the wave enters the lens at **$L_1$** when $t$ = 8.5 μs
- (From Fig. 1.3) the wave leaves the lens at **$L_2$** when $t$ = 14 μs
- Time taken: $\Delta t$ = 14 – 8.5 = 5.5 μs

Calculate the thickness of the lens:




- (From Fig. 1.3) the wave enters the lens at $L_1$ when $t$ = 8.5 μs
- (From Fig. 1.3) the wave leaves the lens at $L_2$ when $t$ = 14 μs
- Time taken: $\Delta t$ = 14 – 8.5 = 5.5 μs

Calculate the thickness of the lens:

- Lens thickness = $\frac{1}{2} c_{lens} \Delta t$ [1 mark]
- Lens thickness = $\frac{1}{2} \times 1670 \times (5.5 \times 10^{-6})$
- Lens thickness = $4.59 \times 10^{-3}$ m = 4.6 mm [1 mark]

(ii) Calculate the thickness of the eye :

- (From Fig. 1.3) the wave enters the eye at A when $t$ = 5.5 μs
- (From Fig. 1.3) the wave meets the back of the eye at B when $t$ = 31.5 μs
- Time taken from **A** to $L_1$: $\Delta t_A$ = 8.5 – 5.5 = 3.0 μs
- Time taken from $L_2$ to **B**: $\Delta t_B$ = 31.5 – 14 = 17.5 μs

Calculate the distances before and after the lens:

- Distance **A** → $L_1$ = $\frac{1}{2} c_{eye} \Delta t_A$
- Distance **A** → $L_1$ = $\frac{1}{2} \times 1520 \times (3.0 \times 10^{-6}) = 2.28 \times 10^{-3}$ m [1 mark]
- Distance $L_2$ → **B** = $\frac{1}{2} c_{eye} \Delta t_B$
- Distance $L_2$ → **B** = $\frac{1}{2} \times 1520 \times (17.5 \times 10^{-6}) = 1.33 \times 10^{-2}$ m [1 mark]

Calculate the thickness of the eye:

- Eye thickness = distance (**A** → $\mathbf{L_1}$) + lens thickness + distance ($\mathbf{L_2}$ → **B**)
- Eye thickness = 0.23 + 0.46 + 1.33 = 2.02 cm [1 mark]

**[Total: 5 marks]**

Take some time to decipher the graph to make sure you understand it before diving into the calculations

At each boundary between media, some of the energy of the ultrasound pulse is **reflected** back to the transducer and the rest is **transmitted.** The reflected pulses are what give rise to the spikes we see on the graph

- The first spike represents the ultrasound entering the cornea at boundary **A**
- The next two spikes represent the boundaries $\mathbf{L_1}$ and $\mathbf{L_2}$, which are the front and rear surfaces of the eye lens
- The final spike is where contact is made with the retina at boundary **B**, i.e. the back of the eye

1d

(d) Calculate the ratio of the intensity of the reflections from A and B:

List the known quantities:

- Acoustic impedance of the eye, $Z_e = 1.52 \times 10^6$ kg m$^{-2}$ s$^{-1}$
- Acoustic impedance of the lens, $Z_l = 1.84 \times 10^6$ kg m$^{-2}$ s$^{-1}$
- Acoustic impedance of the surrounding tissue, $Z_t = 1.69 \times 10^6$ kg m$^{-2}$ s$^{-1}$

Calculate the intensity reflection coefficient at each boundary:

- Intensity reflection coefficient: $\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
- Eye-lens boundary: $\frac{I_R}{I_0} = \frac{(1.84 \times 10^6 - 1.52 \times 10^6)^2}{(1.84 \times 10^6 + 1.52 \times 10^6)^2} = 9.1 \times 10^{-3}$ [1 mark]

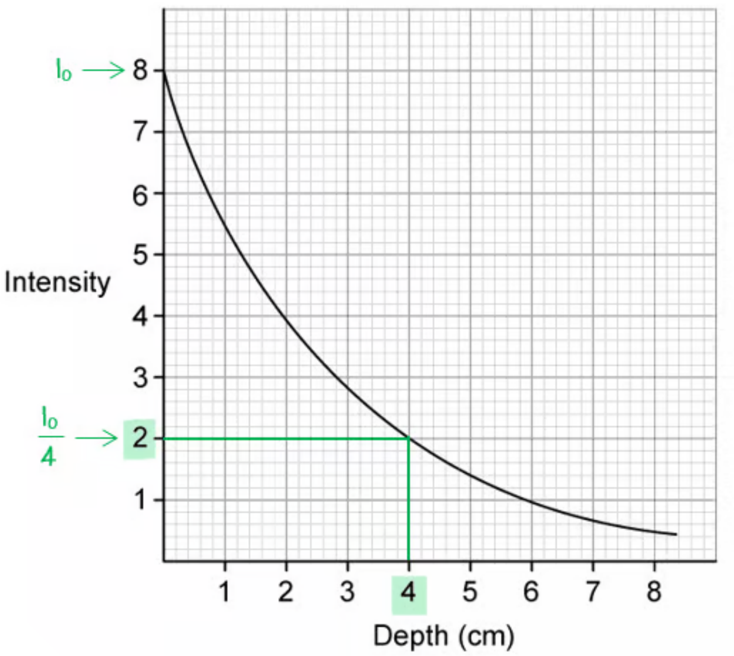
- Eye-tissue boundary: $\frac{I_R}{I_0} = \frac{(1.69 \times 10^6 - 1.52 \times 10^6)^2}{(1.69 \times 10^6 + 1.52 \times 10^6)^2} = 2.8 \times 10^{-3}$ [1 mark]

Determine the total energy lost by reflection between A and B:

- At each boundary, only a small fraction of energy is reflected
- Hence, transmitted energy at B = (0.9909 × 0.9909 × 0.9972) = 0.9791, so 0.021 of that is reflected
- It is reasonable to make the approximation 0.9791 ≈ 1.0
- So the same amount of energy is reflected at both A and B; [1 mark]

Determine the energy lost by attenuation:

- The decrease in intensity must be due to the attenuation of the signal only
- Total distance travelled = 2 × eye thickness = 4 cm




- (From the graph) when depth = 4 cm, the intensity is reduced from 8 to 2, or $I_0 \rightarrow \frac{I_0}{4}$
- Therefore, $\frac{I_A}{I_B} = 4$ [1 mark]

**[Total: 4 marks]**

At first, this question looks more complicated than it really is. When you realise that all the energy loss results from attenuation, rather than reflection, it simplifies the question greatly

Remember that attenuation happens in both directions. It is reasonable to assume that attenuation occurs at the same rate in all tissues, so you don't need to determine attenuation in the lens and eye separately

2a

(a) The energy that would produce the best contrast is:

- 50 keV; [1 mark]

This is because:

- Contrast depends on the difference in attenuation; [1 mark]
- The difference between $\mu$ of bone and muscle is greatest with 50 keV X-rays (hence best contrast); [1 mark]

**[Total: 3 marks]**

2b

(b)

(i) Show that bone absorbs X-rays about eight times more strongly than muscle:

Determine the average values of Z

| substance | main elements | average $Z$ |
|---|---|---|
| muscle | $_{1}H$, $_{6}C$, $_{8}O$ | **5** |
| bone | $_{1}H$, $_{6}C$, $_{8}O$, $_{15}P$, $_{20}Ca$ | **10** |
| contrast media | $_{53}I$, $_{56}Ba$ | **55** |

- Average values of $Z$ calculated; [1 mark]

Determine the ratio of the average values of $Z$:

(hence best contrast); [1 mark]

**[Total: 3 marks]**

2b

(b)

(i) Show that bone absorbs X-rays about eight times more strongly than muscle:

Determine the average values of Z

| substance | main elements | average *Z* |
|---|---|---|
| muscle | $_{1}H,\ _{6}C,\ _{8}O$ | **5** |
| bone | $_{1}H,\ _{6}C,\ _{8}O,\ _{15}P,\ _{20}Ca$ | **10** |
| contrast media | $_{53}I,\ _{56}Ba$ | **55** |

- Average values of *Z* calculated; [1 mark]

Determine the ratio of the average values of *Z*:

- Ratio: $\frac{Z_{bone}}{Z_{muscle}} \approx \frac{10}{5} \approx 2$ [1 mark]

Use the relationship between μ and Z

- Attenuation coefficient: $\mu \propto Z^3$
- Therefore $\frac{\mu_{bone}}{\mu_{muscle}} = 2^3 = 8$ [1 mark]

(ii) Contrast media are made from elements with high values of *Z* because:

- A contrast medium is a substance which absorbs X-rays strongly; [1 mark]
- As X-rays pass through a medium, they interact with electrons and/or nuclei (of atoms); [1 mark]
- Hence, substances with higher numbers of nuclei / electrons will interact with X-rays more, making them more absorbing; [1 mark]

**[Total: 6 marks]**

2c

(c) X-rays with energies less than 20 keV are usually filtered out of the beam because:

- (At energies less than 20 keV) there is more attenuation **OR** (at energies greater than 20 keV) there is less attenuation; [1 mark]
- The lower energy photons will be mostly absorbed (by muscles); [1 mark]
- Hence these (low energy) photons will not contribute to the imaging process; [1 mark]
- Only the higher energy photons will leave the body and be detected by the sensor; [1 mark]

**[Total: 4 marks]**

It's important to realise that almost **all** of the low-energy X-rays would be attenuated inside the body, therefore, these X-rays would

- Never even reach the sensor
- Unnecessarily expose the patient to a higher dose of harmful radiation

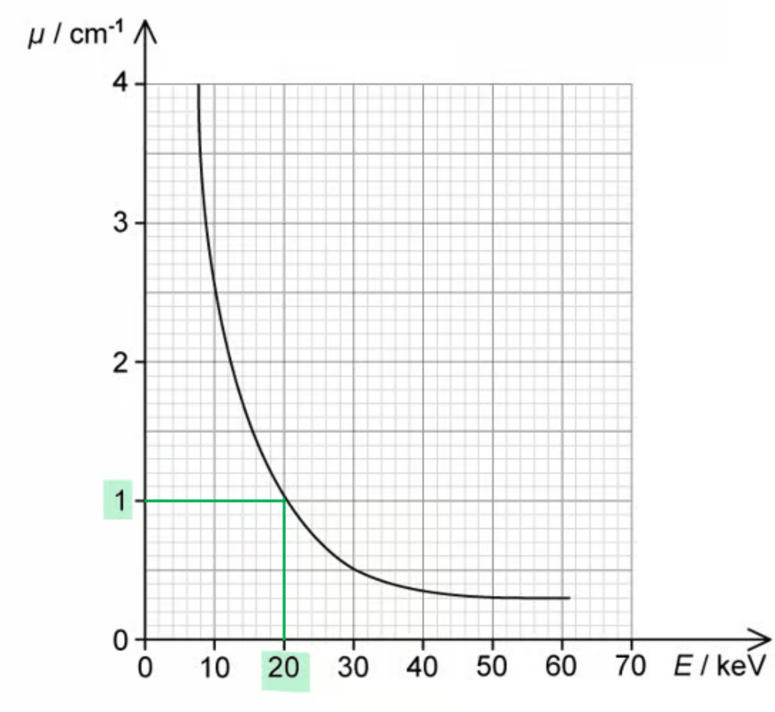
2d

(d) Calculate the ratio of intensities to compare the attenuation produced by 1 cm of bone and 1 cm of muscle:

List the known quantities:

- Thickness of muscle/bone, $x$ = 1 cm

Determine μ for muscle and bone at 20 keV:




- (From graph) when $E = 20$ keV, $\mu_{muscle} = 1.0$ cm$^{-1}$ [1 mark]

- Since $\mu \propto Z^3$ and $\frac{\mu_{bone}}{\mu_{muscle}} \approx 8$ (from **(b)**)

- Therefore $\mu_{bone} = 8 \times 1.0 = 8.0 \text{ cm}^{-1}$ [1 mark]

Use the attenuation equation to determine the ratio of intensities:

- X-ray attenuation: $I = I_0 e^{-\mu x}$

- Ratio of intensities: $\frac{I_b}{I_m} = \frac{e^{-\mu x}}{e^{-\mu x}}$

- Ratio of intensities: $\frac{I_b}{I_m} = \frac{e^{-8.0 \times 1}}{e^{-1.0 \times 1}} = 9.1 \times 10^{-4}$ [1 mark]

**[Total: 3 marks]**

3a

(a)

(i) Calculate the percentage of energy from the ultrasonic transducer that is reflected:

- Acoustic impedance: $Z = \rho c$
- Acoustic impedance of soft tissue: $Z_{st} = 1065 \times 1530 = 1.629 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
- Acoustic impedance of cyst: $Z_c = 1020 \times 1570 = 1.601 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
- Acoustic impedance of tumour: $Z_t = 990 \times 1565 = 1.549 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$

*1 mark for all Z values correct*

Use the intensity reflection coefficient equation:

- Intensity reflection coefficient: $\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$
- Cyst: $\frac{I_R}{I_0} = \frac{(1.629 \times 10^6 - 1.601 \times 10^6)^2}{(1.629 \times 10^6 + 1.601 \times 10^6)^2} = 7.5 \times 10^{-5}$ [1 mark]
- Tumour: $\frac{I_R}{I_0} = \frac{(1.629 \times 10^6 - 1.549 \times 10^6)^2}{(1.629 \times 10^6 + 1.549 \times 10^6)^2} = 6.3 \times 10^{-4}$ [1 mark]

Compare the ratios to the ratio given in the question to identify the medium:

- $\frac{I_R}{I_0} = 7.54 \times 10^{-5} = 7.5 \times 10^{-5} \text{ (2 s.f.)}$ [1 mark]
- Therefore, the unknown medium is a cyst

(ii) Determine the length of the cyst:

List the known quantities:

- Time detected from front of cyst = 27.5 μs
- Time detected from rear of cyst = 43.2 μs
- Speed of ultrasound in cyst, $c$ = 1570 m s$^{-1}$

- Acoustic impedance of soft tissue: $Z_{st} = 1065 \times 1530 = 1.629 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
- Acoustic impedance of cyst: $Z_c = 1020 \times 1570 = 1.601 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$
- Acoustic impedance of tumour: $Z_t = 990 \times 1565 = 1.549 \times 10^6 \text{ kg m}^{-2} \text{ s}^{-1}$

*1 mark for all Z values correct*

Use the intensity reflection coefficient equation:

- Intensity reflection coefficient: $\frac{I_R}{I_0} = \frac{(Z_1 - Z_2)^2}{(Z_1 + Z_2)^2}$

- Cyst: $\frac{I_R}{I_0} = \frac{(1.629 \times 10^6 - 1.601 \times 10^6)^2}{(1.629 \times 10^6 + 1.601 \times 10^6)^2} = 7.5 \times 10^{-5}$ [1 mark]

- Tumour: $\frac{I_R}{I_0} = \frac{(1.629 \times 10^6 - 1.549 \times 10^6)^2}{(1.629 \times 10^6 + 1.549 \times 10^6)^2} = 6.3 \times 10^{-4}$ [1 mark]

Compare the ratios to the ratio given in the question to identify the medium:

- $\frac{I_R}{I_0} = 7.54 \times 10^{-5} = 7.5 \times 10^{-5} \text{ (2 s.f.)}$ [1 mark]
- Therefore, the unknown medium is a cyst

(ii) Determine the length of the cyst:

List the known quantities:

- Time detected from front of cyst = 27.5 μs
- Time detected from rear of cyst = 43.2 μs
- Speed of ultrasound in cyst, $c$ = 1570 m s$^{-1}$

Calculate the length of the cyst:

- Distance travelled by ultrasound: $d = \frac{1}{2} c \Delta t$

- $d = \frac{1}{2} \times 1570 \times (43.2 - 27.5) \times 10^{-6}$ [1 mark]
- $d = 0.01232 \text{ m} = 1.2 \text{ cm}$ [1 mark]

**[Total: 6 marks]**

3b

(b) Calculate the X-ray energy incident on the cyst each second:

List the known quantities:

- Attenuation constant of soft tissue, $\mu = 0.85 \text{ cm}^{-1}$
- Initial intensity of the beam, $I_0 = 4.6 \times 10^3 \text{ W m}^{-2}$
- Length of the cyst, $d = 0.012$ m = 1.2 cm (from **(a)**)
- Distance travelled through the tissue, $x = 2.1$ cm

Calculate the intensity of the X-ray beam at the cyst:

- Intensity of a beam in tissue: $I = I_0 e^{-\mu x}$
- $I = (4.6 \times 10^3) e^{-(0.85 \times 2.1)} = 771.9 \text{ W m}^{-2}$ [1 mark]

Determine the cross-sectional area of the front surface of the cyst:

- Cross-sectional area of the cyst: $A = \pi r^2 = \pi \times \left(\frac{0.012}{2}\right)^2 = 1.13 \times 10^{-4} \text{ m}^2$

  **AND**
- Assumption: the front surface is circular; [1 mark]

Calculate the power and energy incident on the cyst:

- Intensity of a wave: $I = \frac{P}{A} \Rightarrow P = IA$
- Incident power: $P = 771.9 \times (1.13 \times 10^{-4}) = 0.087 \text{ W}$
- Energy transferred each second: $E = Pt = 0.087 \text{ J}$ [1 mark]

**[Total: 3 marks]**

3c

(c)

(i) Determine the attenuation coefficient of the cyst:

List the known quantities:

- Attenuation coefficient of soft tissue, $\mu$ = 0.85 cm$^{-1}$
- Initial intensity of the beam, $I_0$ = 4.6 × 10$^3$ W m$^{-2}$
- Length of the cyst, $d$ = 0.012 m = 1.2 cm (from **(a)**)
- Distance between the tissue and cyst = 2.1 cm
- Total distance = 4.8 cm

Use the attenuation equation:

- X-ray attenuation: $I = I_0\, e^{-\mu x}$
- Final intensity of beam X: $I_X = \left(4.6 \times 10^3\right) e^{-0.85 \times 4.8}$
- Final intensity of beam Y: $I_Y = \left(4.6 \times 10^3\right) \left[\left(e^{-0.85 \times 2.1}\right)\left(e^{-1.2\mu}\right)\left(e^{-0.85 \times (4.8 - 1.2 - 2.1)}\right)\right]$ [1 mark]

Substitute into the ratio given in the question:

- Ratio of the emergent intensities: $\frac{I_Y}{I_X} = 0.94$
- Substitute: $\frac{\left(4.6 \times 10^3\right) \left[\left(e^{-0.85 \times 2.1}\right)\left(e^{-1.2\mu}\right)\left(e^{-0.85 \times (4.8 - 1.2 - 2.1)}\right)\right]}{\left(4.6 \times 10^3\right) e^{-0.85 \times 4.8}} = 0.94$ [1 mark]
- Simplify: $e^{0.85(4.8 - 2.1 - 1.5) - 1.2\mu} = 0.94 \quad \Rightarrow \quad e^{1.02 - 1.2\mu} = 0.94$
- Takes natural logs: $1.02 - 1.2\mu = \ln 0.94 \quad \Rightarrow \quad \mu = \frac{1.02 - \ln 0.94}{1.2}$
- Attenuation coefficient of cyst: $\mu$ = 0.90 cm$^{-1}$ [1 mark]

(ii) Compare the quality of the images that would be obtained from the ultrasound and the

X-ray scans:

For the X-ray image...

- The attenuation coefficients (of the cyst and the surrounding tissue) are too similar / have approximately the same value
  **AND**
- (Hence) no / poor contrast would be obtained between the cyst and surrounding tissue / cyst would not be distinguishable from the surrounding tissue; [1 mark]

For the ultrasound image...

- The acoustic impedances (of the cyst and the surrounding tissue) are similar / have approximately the same value
  **AND**
- (Hence) there will be a high % of transmission of the ultrasound / low % of reflection (so) cyst would be clearly distinguishable from the surrounding tissue; [1 mark]

**[Total: 5 marks]**

3d

(d) A different technique that could be used to image the cyst is:

- A CT scan **OR** PET scan; [1 mark]

An advantage of CT or PET over ultrasound is:

Max **one** from:

- A CT / PET scan will give a detailed 3-dimensional image (of the cyst); [1 mark]
- A CT / PET scan gives better contrast; [1 mark]
- A PET scan can determine metabolic activity; [1 mark]

A disadvantage of CT or PET over ultrasound is:

Max **one** from:

- Both CT / PET scans pose a danger (to the patient) due to high radiation exposure; [1 mark]
- Ultrasound can determine the same information i.e. location / size / shape /

For the ultrasound image...

- The acoustic impedances (of the cyst and the surrounding tissue) are similar / have approximately the same value
  **AND**
- (Hence) there will be a high % of transmission of the ultrasound / low % of reflection (so) cyst would be clearly distinguishable from the surrounding tissue; [1 mark]

**[Total: 5 marks]**

3d

(d) A different technique that could be used to image the cyst is:

- A CT scan **OR** PET scan; [1 mark]

An advantage of CT or PET over ultrasound is:

Max **one** from:

- A CT / PET scan will give a detailed 3-dimensional image (of the cyst); [1 mark]
- A CT / PET scan gives better contrast; [1 mark]
- A PET scan can determine metabolic activity; [1 mark]

A disadvantage of CT or PET over ultrasound is:

Max **one** from:

- Both CT / PET scans pose a danger (to the patient) due to high radiation exposure; [1 mark]
- Ultrasound can determine the same information i.e. location / size / shape / composition (solid or fluid-filled) just as well as CT / PET with zero radiation risk; [1 mark]

**[Total: 3 marks]**

Make sure to use specific and detailed language such as 'better contrast' or 'metabolic activity'

Saying 'more detail' or 'clearer image' is too vague and will not be awarded a mark